

EXACT UNIVERSALITY FOR PROJECTIVE REALIZATION SPACES OF ORIENTED MATROIDS

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ABSTRACT. We prove that every nonempty basic semialgebraic set defined over the rationals by polynomial equations and strict inequalities is rationally equivalent to the labeled projective realization space of a finite oriented matroid. Both maps have rational coefficients and are defined at every real point. In particular, no stable equivalence or family of auxiliary choices is needed. We encode arithmetic on an ordered projective line and realize each scalar relation by four labeled points in an independent projective plane through that line. Every maximal minor then reduces to two line directions, while one projective normalization removes all placement parameters. The rank grows with the number of relations; a universal rank bound remains open. We also prove exact universality for labeled projective realization spaces of convex polytopes of unrestricted dimension. As a proof of concept, we construct a rank-98 oriented matroid on 484 elements whose projective realization space is rationally equivalent to the circle.

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For more information, see: martinwintermath.github.io/ai-math.html

1. INTRODUCTION

An oriented matroid records the signs of the maximal minors of a vector configuration. It remembers incidences and orientations, but forgets the actual coordinates. The resulting realization space measures precisely how much geometry remains after this combinatorial information has been prescribed. How complicated can that space be?

Mnëv’s universality theorem answers this question up to stable equivalence: already oriented matroids of rank three have realization spaces with the complexity of arbitrary primary basic semialgebraic sets [3]. Stable equivalence preserves substantial topological and arithmetic information. It nevertheless permits

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auxiliary parameters, and hence does not identify the realization space with the original set. If we want to prescribe the space itself, rather than its stable-equivalence class, those parameters become the main issue.

We shall work with labeled realizations modulo projective transformations. A labeled realization of a rank- r oriented matroid M on N elements is a full-row-rank matrix $V \in \mathbb{R}^{r \times N}$ with its specified maximal-minor signs, up to simultaneous global sign. Two realizations are projectively equivalent if $W = AVD$, with $A \in \text{GL}_r(\mathbb{R})$ and D a positive diagonal matrix. We write $\mathcal{R}_{\text{proj}}(M)$ for the corresponding quotient. Throughout, a chirotope is understood up to simultaneous global sign; thus a row transformation of negative determinant is allowed, whereas individual reorientations are not. Standard oriented-matroid terminology follows [2].

For semialgebraic sets $X \subseteq \mathbb{R}^a$ and $Y \subseteq \mathbb{R}^b$, we call a bijection $f: X \rightarrow Y$ an *exact rational equivalence* if both f and f^{-1} are restrictions of rational maps over $\mathbb{Q}$, with no denominator vanishing at any real point of their respective domains. We write $X \cong_{\mathbb{Q}} Y$. This is an equivalence of real loci, and in particular a semialgebraic homeomorphism. It need not extend to an isomorphism of complex affine schemes. For a projective realization space, we use an explicitly specified normalization chart; the construction below provides a global chart of this kind.

Theorem 1.1 (Exact universality). *Let*

$$S = \{x \in \mathbb{R}^n \mid F_1(x) = \cdots = F_m(x) = 0, \quad g_1(x) > 0, \dots, g_s(x) > 0\},$$

where all polynomials have rational coefficients, and suppose $S \neq \emptyset$. There is an effectively constructible finite realizable oriented matroid M , without loops or coloops, such that

$$S \cong_{\mathbb{Q}} \mathcal{R}_{\text{proj}}(M).$$

The realization space admits a unique normalized representative for each projective class, and the equivalence and its inverse are rational in these normalized coordinates.

In particular, every nonempty real-algebraic set defined over $\mathbb{Q}$ occurs exactly, without assumptions on compactness, dimension, connectedness or singularities. We obtain the same conclusion for convex polytopes. For a labeled convex polytope P , let $\mathcal{R}_{\text{proj}}(P)$ denote the space of convex realizations of its labeled face lattice, modulo projectivities defined on the whole polytope and preserving its vertex labels.

Theorem 1.2 (Exact universality for convex polytopes). *Let S be a nonempty set defined by rational polynomial equations and strict inequalities as in Theorem 1.1. There is an effectively constructible convex polytope P such that*

$$S \cong_{\mathbb{Q}} \mathcal{R}_{\text{proj}}(P).$$

In a global rational normalization of the projective realization space, the equivalence and its inverse are defined at every real point.

We prove Theorem 1.2 in Section 5 by transferring the oriented-matroid result to polytopes. No dimension bound is asserted.

The tradeoff is the rank. Unlike classical universality, our construction does not give a rank bound independent of the input. We use one transverse direction for each arithmetic relation, and the rank grows

accordingly. Whether exact universality can be achieved with universally bounded rank remains open; in particular, we do not obtain an exact rank-three theorem.

The proof rests on two observations. First, arithmetic can be encoded by ordered lists of labeled points on one projective line. We call these lists clusters, and define them precisely in [Section 2.4](#). Identifying successive endpoints fixes their translations. Second, a quadrilateral implementing one relation has no projective modulus relative to that line. The difficulty is that several quadrilaterals in the same plane have independent relative placements. Classical constructions control their incidences by allowing suitable open choices of those placements [\[4, §11.7\]](#). We instead put the quadrilaterals in independent planes through the common line. A single ambient projectivity then normalizes all of them simultaneously.

It remains to check that this change does not introduce uncontrolled determinant signs. Here the increasing rank helps a second time: in a maximal minor, all transverse rows can be eliminated, leaving only two directions on the common line. Its prescribed order therefore controls every minor, including those meeting several different quadrilaterals. This is the structural reason why the construction is exact.

2. ARITHMETIC ON AN ORDERED LINE

We construct a space $\mathcal{W}$ of ordered labeled line configurations exactly rationally equivalent to S . This converts the defining polynomial conditions into scalar relations that we can subsequently realize by an oriented matroid. We first define the finite arithmetic data, then specify their point labels and geometric constraints.

2.1. Arithmetic circuits.

Definition 2.1. An *arithmetic circuit* consists of a finite set I of input labels, a distinguished unit label $1 \notin I$, an ordered list of assignments

$$t_k = a_k \circ_k b_k \quad (1 \leq k \leq h),$$

and a finite list of *terminal pairs* (L_j, R_j) , each prescribing equality of its two evaluated values. Each assignment, called a *gate*, has $\circ_k \in \{+, \cdot\}$, an output label $t_k \notin I \cup \{1, t_1, \dots, t_{k-1}\}$, and operand labels $a_k, b_k \in I \cup \{1, t_1, \dots, t_{k-1}\}$. The terminal labels belong to $I \cup \{t_1, \dots, t_h\}$, excluding the unit label.

The *circuit variables* are the labels in $T = I \cup \{t_1, \dots, t_h\}$. An *evaluation* assigns numbers ν_t to these labels, with $\nu_1 = 1$ and $\nu_{t_k} = \nu_{a_k} \circ_k \nu_{b_k}$. We call an evaluation *admissible* if $\nu_t > 1$ for all $t \in T$ and $\nu_{L_j} = \nu_{R_j}$ for all terminal pairs.

Thus a gate is one specified arithmetic assignment, and a circuit is their finite ordered list together with its inputs and terminal equations. Labels are distinct even when an evaluation assigns them equal numbers.

2.2. Positive coordinates. For each original coordinate x_i , introduce

$$u_i = x_i^2 + 2, \quad v_i = x_i^2 + x_i + 3.$$

Both numbers exceed 1, and $x_i = v_i - u_i - 1$. Conversely, the equation

$$u_i = (v_i - u_i - 1)^2 + 2$$

recovers exactly these expressions: after setting $x_i = v_i - u_i - 1$, it also gives $v_i = x_i^2 + x_i + 3 \geq 11/4$. Thus this change of variables is polynomially invertible on its image.

For each strict inequality $g_j(x) > 0$, introduce $z_j > 1$ and impose

$$g_j(x)(z_j - 1) = 1.$$

The new variable is uniquely determined, namely $z_j = 1 + 1/g_j(x)$; conversely its strict lower bound forces $g_j(x) > 0$. This is an exact rational equivalence. The positivity restriction on z_j remains part of the defining conditions.

Substitute the positive coordinates in all equations and clear rational coefficient denominators. Splitting each polynomial into its positive and negative monomials writes the equation as $P = Q$, with nonnegative integer coefficients. Replace it by $P + 2 = Q + 2$. We evaluate both sides by an arithmetic circuit in the sense of [Definition 2.1](#), using $+$, multiplication, and the unit label 1. We omit multiplication by 1; every input and every output label then has value greater than 1. A constant expression equal to 1 uses the unit label itself. The terminal pairs express the resulting equations $P + 2 = Q + 2$.

The resulting positive-coordinate system is a basic semialgebraic set exactly rationally equivalent to S . If the original set has no strict inequalities, its polynomial equations alone force all the required positivity. With strict inequalities, we retain the positivity conditions throughout.

2.3. Point labels and quadrilateral relations. To a circuit with variable-label set T , associate the finite set

$$\Sigma = \{0, 1, -1\} \cup \{t, -t, t^{-1} \mid t \in T\}.$$

We call its elements *symbols*. The union is understood as disjoint: all the displayed labels are stipulated distinct. Here $t, -t, t^{-1}$ are three distinct formal labels, not numerical expressions to be evaluated before a configuration is chosen. For an admissible circuit evaluation, their intended values are $\nu_t, -\nu_t, 1/\nu_t$, respectively; the intended values of $0, 1, -1$ are their usual numerical values. The relations below will force this interpretation. For convenience, we also write $1/t$ for the symbol t^{-1} .

For a real coordinate τ on a projective line write $P_\tau = [\tau : 1]$ and $P_\infty = [1 : 0]$. For finite a, b, c, d define

$$Q(a, b, c, d; f) = \frac{(a-d)(c-f)}{(a-f)(c-b)} \quad (f \in \mathbb{R}), \quad Q(a, b, c, d; \infty) = \frac{a-d}{c-b}.$$

We use only relations $Q = 1$ with nonzero displayed denominators. They are invariant under simultaneous translation of all finite arguments. The following relations impose the required arithmetic. A Q -expression in symbols means Q applied to the coordinates assigned to those point labels:

relation	constraint
$Q(0, 0, -1, 1; \infty) = 1$	initial unit scale
$Q(0, 0, -t, t; \infty) = 1$	negative of t
$Q(1, 1, 1/t, t; 0) = 1$	reciprocal of t
$Q(0, u, -t, w; \infty) = 1$	$w = t + u$
$Q(1, u, 1/t, w; 0) = 1$	$w = tu$
$Q(0, 0, -t, u; \infty) = 1$	$t = u$

For a variable label t , the negative and reciprocal relations force the points labeled $-t$ and $1/t$ to have the intended values relative to the local origin and unit. Once the local origin and unit are normalized and the negative and reciprocal relations hold, the addition and multiplication expressions simplify to $w/(t+u)$ and

$(1 - w)/(1 - tu)$. The multiplication denominator does not vanish because multiplication gates have both inputs greater than 1.

2.4. Clusters and linking relations. A *cluster* is a finite ordered list of distinct symbols in Σ , containing 0, together with specified $Q = 1$ relations among them and the common label ∞ . An *occurrence* is a pair (i, σ) consisting of a cluster index and a symbol in its list. In a geometric realization, each occurrence is a labeled point on the common projective line, and the list prescribes their increasing affine order. The occurrence of 0 is the cluster's *local origin*. After choosing an affine coordinate on that line, the *local coordinate* of any occurrence is its coordinate minus that of the local origin. We write σ^i for this occurrence and, in arithmetic expressions, for its affine coordinate.

The *left endpoint* and *right endpoint* of a cluster are its first and last occurrences, respectively, or the corresponding labeled points in a realization. They are the points of minimum and maximum local coordinate. The closed affine interval between them is the interval occupied by the cluster; its interior means the corresponding open interval. The common point P_∞ belongs to neither the list nor this interval.

Throughout these lists, t^{-1} (also written $1/t$) remains a formal symbol; its local coordinate is forced to be a reciprocal by the relations. The same symbol in different clusters initially gives different point labels. Only the endpoint identifications specified below merge occurrences. Thus a cluster is explicit finite labeling and order data for part of a line configuration. We use the following lists, ordered by the intended local values:

cluster	local points
initial	$-1, 0, 1$
variable t	$-t, 0, 1/t, 1, t$
addition $t + u = w$	$-t, 0, 1, u, w$
multiplication $tu = w$	$0, 1/t, 1, u, w$
equality $t = u$	$-t, 0, 1, u$

For addition, t or u may be 1. If $u = 1$, its occurrence and the displayed unit point are one label; all remaining inequalities are strict. Multiplication gates have no unit input. The initial cluster carries its unit relation, each variable cluster carries its negative and reciprocal relations, and each gate or equality cluster carries its indicated relation. We call a relation imposed within one cluster a *local relation*. This use of quadrilateral relations is related to the arithmetic encoding in [4, §§11.2–11.4].

Order the clusters as $0, \dots, C - 1$, beginning with the initial cluster. Write m_i, M_i for the minimum and maximum local values of cluster i , and s_i for its *offset*, the global affine coordinate of its local origin. Identify its right endpoint with the left endpoint of the next cluster. Set $s_0 = 0$. The offsets are forced by

$$(2.1) \quad s_{i+1} = s_i + M_i - m_{i+1}.$$

Every cluster has positive length, $M_i > 0$ and $m_i \leq 0$. Thus the interiors of consecutive clusters are disjoint and all global origins are strictly increasing. The origin of a multiplication cluster is its left endpoint.

Lemma 2.2. *In the ordered configuration defined by (2.1), two distinct finite occurrences have the same global coordinate if and only if they are the right endpoint of one cluster and the left endpoint of the immediately following cluster. After these identifications, all remaining line labels, including P_∞ , represent distinct points.*

Proof. Write $\ell_i = s_i + m_i$ and $r_i = s_i + M_i$ for the global endpoint coordinates. The recurrence gives $r_i = \ell_{i+1}$, and the strict local order gives $\ell_i < r_i$. Consequently adjacent cluster intervals meet only at their prescribed common endpoint, while $r_i < \ell_j$ for $j \geq i + 2$. Within one cluster all occurrences are distinct. These observations exhaust all possible coincidences; P_∞ is distinct from every finite point. $\square$

Repeated symbols in $\Sigma \setminus \{0\}$ must have the same local value. If w^i, w^j are occurrences of the same such symbol in clusters i, j , with origins $0^i, 0^j$, impose

$$(2.2) \quad Q(0^i, w^j, 0^j, w^i; \infty) = 1.$$

This says exactly $w^i - 0^i = w^j - 0^j$. We call (2.2) a *linking relation*. For each symbol in $\Sigma \setminus \{0\}$ we impose one such relation from every later occurrence to its first occurrence. The symbols $t, -t, t^{-1}$ and $1, -1$ are treated separately. The zero occurrences are not linked.

All the relations satisfy

$$(2.3) \quad a \neq d, \quad c \neq b, \quad a \neq c, \quad b \neq d,$$

and, for finite f , also $a \neq f$ and $c \neq f$. For local relations this follows from the prescribed order. For a link, the first two differences are the common nonzero local value, and the two origins are distinct. Finally, a shared endpoint has a positive local label in its left cluster and a negative or zero local label in its right cluster. It therefore cannot identify two copies of the same nonzero formal symbol, which gives $b \neq d$.

The arguments a, b, c, d of a relation are not required to be pairwise distinct. For example, the unit and negative relations have $a = b = 0$. Only the inequalities in (2.3), and the two additional inequalities for finite f , are required to obtain a nondegenerate quadrilateral. The label identifications are fixed in advance by Lemma 2.2.

Let $\mathcal{W}$ be the space of labeled ordered line configurations satisfying all these relations, normalized by the initial 0, 1 and the common ∞ . All labels refer to global positions after the specified endpoint identifications. Equal numerical values in different local coordinates are not identified unless they are the prescribed common endpoints.

Lemma 2.3. *The circuit construction gives an exact rational equivalence $S \cong_{\mathbb{Q}} \mathcal{W}$. Every marked coordinate is uniquely determined by the input, and the original coordinates are affine-linear functions of differences of marked coordinates.*

Proof. Given an input, evaluate the circuit, its negatives and reciprocals, and then the offsets by (2.1). This produces the required ordered line configuration. The only denominators introduced are positive circuit-variable values and the nonvanishing input inequality polynomials.

Conversely, in any element of $\mathcal{W}$, subtract each local origin from its marked coordinates. Linking makes repeated symbols agree. The initial relation gives the negative unit, the variable relations give negatives and reciprocals, and the gate relations enforce circuit evaluation. The terminal relations impose the original equations. The strict order in the variable clusters gives every circuit variable a value greater than 1, including the reciprocal variables for the original inequalities. The preprocessing therefore recovers a unique point of S .

The glued endpoints recover (2.1), so no cluster translation is free. Original coordinates are recovered by $x_i = v_i - u_i - 1$, with each variable value read as its marked coordinate minus its local origin. The maps are inverse and have rational coefficients. $\square$

3. INDEPENDENT QUADRILATERAL PLANES

We construct a vector configuration from each element of $\mathcal{W}$ and prove that all these configurations realize the same oriented matroid M . This turns the scalar constraints of Section 2 into one finite sign pattern. We realize each relation in a separate projective plane through the common line; Section 4 will establish the converse correspondence.

3.1. One relation. A *quadrilateral* will mean four labeled points A, B, C, D in one projective plane, with no collinear triple. For two such points, AB denotes their joining projective line, and similarly for the other pairs. Place the common line L at $z = 0$ in $\mathbb{P}(\mathbb{R}^3)$, with $P_t = [t : 1 : 0]$ and $P_\infty = [1 : 0 : 0]$. Here and below, the arguments a, b, c, d, f denote the actual affine coordinates assigned to the point labels of a relation. For $Q(a, b, c, d; f) = 1$, set $h = 1/(a - d)$, $k = 1/(c - b)$ and choose representatives

$$(3.1) \quad A = (0, 0, 1), \quad B = (1, 0, 1), \quad C = (ah, h, 1), \quad D = (ck, k, 1).$$

Their joining lines meet the common line as follows:

joining line	AB	AC	BC	AD	BD	CD
intersection	P_∞	P_a	P_d	P_c	P_b	P_f

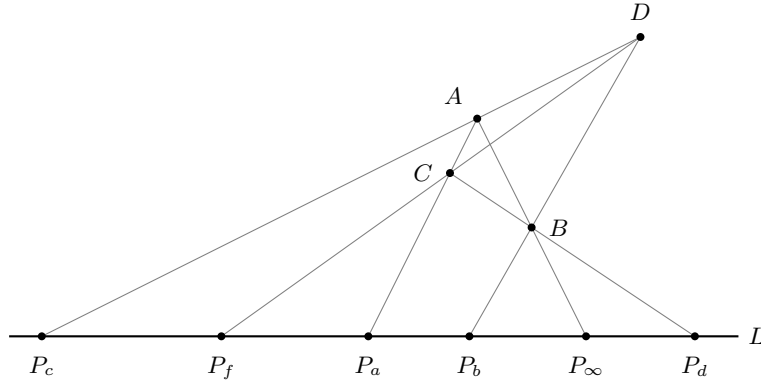


Figure 1. The six joining lines of a quadrilateral meet the common line L at the prescribed labels. We draw L in a finite affine chart; its distinguished point P_∞ is the point used as infinity in (3.1). The picture shows distinct intersections, although some intersection points coincide in the actual relations.

The first five incidences follow directly from (3.1). For finite f , the last is $h(a - f) = k(c - f)$, equivalent to $Q = 1$. For $f = \infty$, it is $h = k$. The triple determinants in the order ABC, ABD, ACD, BCD are

$$h, \quad k, \quad hk(a - c), \quad hk(d - b).$$

By (2.3) they are nonzero.

Conversely, once $A = (0, 0, 1)$ and $B = (1, 0, 1)$ are normalized, these incidences force C, D uniquely. Indeed, if $C = (\xi, \eta, 1)$, then $AC \cap L = P_a$ and $BC \cap L = P_d$ give $\xi = a\eta$ and $\xi - 1 = d\eta$, hence $\eta = h$. The argument for D is identical. Thus a single quadrilateral has no remaining placement parameter in this normalization.

3.2. The full configuration. Let K be the number of local and linking relations, and n_L the number of distinct line labels, including P_∞ . In

$$E = \mathbb{R}^{K+2} = U \oplus \text{span}(e_1, \dots, e_K), \quad U = \text{span}(e_x, e_y),$$

represent the line labels by $P_t = te_x + e_y$ and $P_\infty = e_x$. For relation i , put

$$(3.2) \quad \begin{aligned} A_i &= e_i, & B_i &= e_x + e_i, \\ C_i &= h_i(a_i e_x + e_y) + e_i, & D_i &= k_i(c_i e_x + e_y) + e_i. \end{aligned}$$

All four columns lie in $U + \mathbb{R}e_i$, and every transverse coordinate in that quadrilateral equals 1. The total rank is $r = K + 2$, as witnessed by $P_\infty, P_0, A_1, \dots, A_K$.

Lemma 3.1. *Every maximal minor of the configuration (3.2), together with its line columns, has constant sign on $\mathcal{W}$. Thus the configurations realize one labeled oriented matroid M of rank $K + 2$ on $n_L + 4K$ elements.*

Proof. Choose $K + 2$ columns. If a quadrilateral is absent, its transverse row vanishes and the minor is zero. Otherwise let $n_i \geq 1$ count the selected columns of quadrilateral i , and let ℓ count selected line columns. Then

$$\ell + \sum_{i=1}^K (n_i - 1) = 2.$$

Choose one selected column per quadrilateral as pivot and subtract it from the other selected columns of that quadrilateral. The transverse rows now occur only in the pivots. Expanding along them leaves, up to a fixed permutation sign, a 2×2 determinant of two vectors in U .

Those vectors are line columns or differences of quadrilateral columns. All pair-differences are determined by

$$\begin{aligned} B - A &= P_\infty, & C - A &= hP_a, & C - B &= hP_d, \\ D - A &= kP_c, & D - B &= kP_b. \end{aligned}$$

For finite f we also have

$$C - D = (h - k)P_f, \quad h - k = \frac{k(c - a)}{a - f} \neq 0;$$

for $f = \infty$ we have $C - D = h(a - c)P_\infty \neq 0$. The factors have fixed signs by the prescribed global line order. The surviving determinant is consequently a product of such fixed-sign factors and one of

$$\det(P_s, P_t) = s - t, \quad \det(P_\infty, P_t) = 1.$$

Its sign or zero status is fixed as well.

For completeness, the four possible selections and their reduced determinants are listed below. Write X_i, Y_i, Z_i for distinct selected vertices of quadrilateral i , and $\Delta_i(X_i, Y_i) = Y_i - X_i \in U$. Each row includes

exactly one column from every quadrilateral not otherwise specified. The displayed determinant agrees with the maximal minor up to its fixed column-permutation sign.

selection beyond those single columns	reduced determinant
P_s, P_t	$\det(P_s, P_t)$
P_s, X_i, Y_i	$\det(P_s, \Delta_i(X_i, Y_i))$
$X_i, Y_i, X_j, Y_j \quad (i \neq j)$	$\det(\Delta_i(X_i, Y_i), \Delta_j(X_j, Y_j))$
X_i, Y_i, Z_i	$\det(\Delta_i(X_i, Y_i), \Delta_i(X_i, Z_i))$

Every difference is a fixed-sign scalar multiple of a prescribed line direction, by the preceding formulas. The table therefore covers all cases, including three selected columns from one quadrilateral. Equal surviving directions give an identically zero minor. $\square$

The point is that no determinant involving an uncontrolled relative placement survives. After the transverse expansion, the whole calculation takes place on the common line.

Remark 3.2. The sign pattern can be computed from the ordered labels and the pair-difference formulas. We do not need to choose a rational point of S , which need not exist. Nonemptiness supplies a real realization and hence verifies that the computed pattern is an oriented matroid.

4. A GLOBAL PROJECTIVE NORMALIZATION

We prove $\mathcal{R}_{\text{proj}}(M) \cong_{\mathbb{Q}} \mathcal{W}$, which completes the oriented-matroid universality theorem when combined with [Lemma 2.3](#). The preceding section constructs a realization from every element of $\mathcal{W}$. We now recover the line configuration from every realization of M and show that a projective class has a unique normalized representative, using only positive column rescalings.

Proposition 4.1. *Every realization of M has a unique representative of the form (3.2), with normalized line data in $\mathcal{W}$. The normalization is rational with denominators nonzero on the entire realization space. Consequently $\mathcal{R}_{\text{proj}}(M) \cong_{\mathbb{Q}} \mathcal{W}$.*

Proof. The underlying matroid is determined by its bases. It therefore determines the rank of every subset, since every independent subset extends to a basis. In the reference configuration the line labels have rank two, their union with each quadrilateral has rank three, and the triples expressing the six incidences with joining lines have rank two. No triple of quadrilateral vertices is dependent. These rank conditions hold in every realization of M .

We give the normalization explicitly. All matrices in the following steps use the fixed label order.

(1) *Basis coordinates.* Let G be the square matrix with ordered columns

$$\mathcal{B} = (P_{\infty}, P_0, A_1, \dots, A_K),$$

and compute the basis coordinates $Z = G^{-1}V$. Write $D_0 = \det G$. For $j \in \{x, y, 1, \dots, K\}$, indexed in this basis order, let $D_j(v)$ be the determinant obtained by replacing the corresponding column of G by a column v in the same position. Cramer's rule gives $Z_j(v) = D_j(v)/D_0$. The basis is independent in M , so D_0 never vanishes.

In these coordinates write $P_1 = \alpha e_x + \beta e_y$ and $B_i = \gamma_i e_x + \delta_i e_i$. The supports follow from the matroid incidences, and the coefficients $\alpha, \beta, \gamma_i, \delta_i$ are positive: the corresponding determinant ratios have the same signs as in (3.2).

(2) *Row transformation and positive representatives.* Set

$$\lambda_y = \alpha/\beta, \quad \lambda_i = \gamma_i/\delta_i, \quad R = \text{diag}(1, \lambda_y, \lambda_1, \dots, \lambda_K).$$

Apply R to Z ; the combined row transformation from the original matrix is RG^{-1} . We then divide each transformed column by the positive number in the following table. Here $Z_y(v)$ and $Z_i(v)$ are coefficients before applying R .

column	positive divisor
P_∞	1
P_0	λ_y
P_1	α
A_i	λ_i
B_i	γ_i
other finite line column v	$\lambda_y Z_y(v)$
C_i, D_i (column v)	$\lambda_i Z_i(v)$

These divisions are changes of homogeneous representatives, hence positive column rescalings. They give

$$(4.1) \quad \begin{aligned} P_\infty &= e_x, & P_0 &= e_y, & P_1 &= e_x + e_y, \\ A_i &= e_i, & B_i &= e_x + e_i & (1 \leq i \leq K). \end{aligned}$$

A projectivity preserving the labeled basis rays is diagonal. Preserving P_1 and every B_i forces all its diagonal entries to agree, so the residual projectivity is the identity.

We record why every division is valid everywhere. The required nonzero basis-replacement determinants are

coefficients	determinants
α, β	$D_x(P_1), D_y(P_1)$
γ_i, δ_i	$D_x(B_i), D_i(B_i)$
$Z_y(v)$ for a finite line column v	$D_y(v)$
$Z_i(C_i), Z_i(D_i)$	$D_i(C_i), D_i(D_i)$.

Each is a basis minor nonzero in the reference configuration, and therefore nonzero in every realization of M . Its ratio to D_0 is positive in that reference configuration and hence throughout the realization space. This verifies positivity of all the table entries as well as the absence of poles; no generic-position restriction is used.

(3) *Recovering the relations.* Every normalized finite line column is $P_t = te_x + e_y$. The minors formed by two line columns and $A_1, \dots, A_K$ recover their pairwise order, so these columns have the prescribed strict global order.

Each normalized C_i, D_i lies in $U + \mathbb{R}e_i$ and has e_i coefficient 1. The incidences $A_i C_i \cap L = P_{a_i}$ and $B_i C_i \cap L = P_{d_i}$ force the displayed formula for C_i ; the other two incidences force D_i . Finally, $C_i D_i \cap L = P_{f_i}$ is precisely $Q = 1$. The normalized line data therefore lie in $\mathcal{W}$.

Conversely, every element of $\mathcal{W}$ gives the displayed representative and realizes M by [Lemma 3.1](#). The formulas force every column, so the representative is unique. Its reconstruction and the normalization use only matrix inverses, ratios of nonzero coefficients and the nonzero line differences in [\(2.3\)](#). Both directions are rational over $\mathbb{Q}$ with no real poles. $\square$

Proof of [Theorem 1.1](#). Combine [Lemma 2.3](#) and [Proposition 4.1](#). All columns are nonzero. To see that there is no coloop, delete an arbitrary column and choose two remaining distinct line labels and one remaining vertex from each quadrilateral. Their determinant is, up to sign, the nonzero determinant of the two line columns, since each chosen quadrilateral column has transverse coordinate 1. The rank therefore remains $K + 2$ after any deletion. $\square$

For reference, the construction has the size formulas

$$(4.2) \quad \text{rank}(M) = K + 2, \quad |M| = n_L + 4K.$$

The equivalence can be viewed even more concretely. Original coordinates are affine-linear functions of the normalized line coordinates. Adjoining these uniquely recovered coordinates identifies the normalized space with the graph of rational functions over S , with no poles on S . This graph description adds no choice parameter.

Remark 4.2. If S is real-algebraic, this graph is real-algebraic as well: write each auxiliary coordinate as $y_j = p_j(x)/q_j(x)$, with q_j nonzero on the entire real locus of S , and impose the original equations and $q_j(x)y_j - p_j(x) = 0$. All sign conditions are then automatic. For inputs with strict inequalities we retain those inequalities; the graph is basic semialgebraic and locally an open part of a real-algebraic set. Exact rational equivalence does not require global algebraicity of that graph.

5. UNIVERSALITY FOR CONVEX POLYTOPES

We prove [Theorem 1.2](#): the labeled projective realization space of a convex polytope can be prescribed exactly as in the oriented-matroid theorem. The full Lawrence construction supplies the polytope. We shall show that its projective realization space is exactly rationally equivalent to that of the original oriented matroid, so this transfer introduces no additional parameters. For the Lawrence construction and its combinatorial properties, see [\[1\]](#); for the realization-space correspondence and reconstruction, see [\[4, §3.3\]](#).

The geometric transfer is standard: every realization of a Lawrence polytope admits the corresponding projective matrix normal form [\[1, Theorem 2.1\]](#). Our exact-equivalence statement requires us to check that the remaining projective freedom agrees precisely with the oriented-matroid quotient, that the normalization is rational with no real poles, and that its projectivities are defined on the whole convex polytope. A homotopy equivalence of realization spaces, as in [\[1, Theorem 5.4\]](#), does not by itself supply these conclusions. The argument below makes these points explicit for any loopless, coloopless realizable oriented matroid; it uses no special feature of our arithmetic construction. Readers interested primarily in oriented matroids may proceed directly to [Section 6](#), while readers interested in exact universality for polytopes will find the required quotient and normalization checks here.

For a general oriented matroid, a global rational normalization can be obtained as follows. Normalize a fixed basis to I_r , so the representing matrix is $[I_r \mid X]$. Basis-replacement minors fix the support and signs

of X . The remaining action is positive diagonal row and column scaling. The bipartite support graph has one vertex for each row and each nonbasis column, with an edge (a, j) precisely when $X_{aj} \neq 0$. Choose a spanning forest in this fixed graph, and normalize its edge entries to their fixed signs $\varepsilon_{aj} \in \{+1, -1\}$. More explicitly, give one vertex per component scaling 1 and propagate the equations $\rho_a \kappa_j X_{aj} = \varepsilon_{aj}$ along the forest. Each newly determined scaling is positive and rational: it is ε_{aj} divided by the product of the already known scaling and X_{aj} . All denominators are nonzero on the entire realization space. A residual scaling multiplies all row factors in a component by one number and all column factors by its reciprocal, so it acts trivially on every entry of that component. The basis columns are rescaled reciprocally to the row factors, keeping them equal to I_r . We obtain a unique normalized matrix. Thus the projective quotient can be represented by a global rational chart also when we do not have the more convenient normalization (4.1).

Let $V = (v_1, \dots, v_N)$ be a rank- r realization of a loopless oriented matroid M . Its *full Lawrence configuration* is

$$(5.1) \quad \Lambda(V) = \begin{pmatrix} V & 0 \\ I_N & I_N \end{pmatrix}, \quad p_i = (v_i, e_i), \quad q_i = (0, e_i).$$

The columns lie in the affine hyperplane $\sum_i t_i = 1$. Their convex hull is the *full Lawrence polytope*, denoted $\Lambda(M)$ when only its labeled combinatorial type is intended. It has dimension $d = r + N - 1$ and $2N$ vertices. Indeed, the q_i span the bottom coordinates, and the differences $p_i - q_i$ recover the rank- r top block. Each pair is an exposed nondegenerate segment: set its bottom coordinate functional to zero and all other bottom coordinate functionals to positive values. Its endpoints are therefore vertices.

Proposition 5.1. *If M has no loops or coloops, the full Lawrence construction induces an exact rational equivalence*

$$\mathcal{R}_{\text{proj}}(M) \cong_{\mathbb{Q}} \mathcal{R}_{\text{proj}}(\Lambda(M)),$$

where the polytope realizations and their projective equivalences preserve all vertex labels and the convexity chart.

Proof. We prove three statements: the labeled face lattice records M ; every realization of that face lattice has the form (5.1) after a projectivity; and the remaining projective freedom is precisely the equivalence on V .

(1) *Faces and covectors.* A homogeneous supporting functional (a, β) has values $a(v_i) + \beta_i$ and β_i on p_i, q_i . It is nonnegative on all columns precisely when $\beta_i \geq \max(0, -a(v_i))$. A face meeting every pair has equality in all these bounds, and its intersection with pair i is

$\text{sgn } a(v_i)$	+	−	0
face vertices	q_i	p_i	p_i, q_i .

Hence these faces recover all signed covectors of M , with the zero covector corresponding to the whole polytope. More generally, for any covector we may choose which pairs a face omits by setting their β_i strictly above the bound, leaving equality on all present pairs. Conversely every face has this description. Thus the covectors determine the entire labeled face lattice, and that lattice recovers M .

(2) *Recovering the Lawrence form.* The vertices outside pair i form a facet F_i . The functional t_i exposes it, and its homogeneous rank is $r + N - 1$: deleting v_i preserves rank r because i is not a coloop. Put

$R_0 = r + N$. We first specify facet functionals rationally in an arbitrary realization of the same labeled face lattice.

For each F_i , fix a maximal flag $E_{i,0} \subset \cdots \subset E_{i,R_0-2} = F_i$, with $\dim E_{i,k} = k$. Choose a vertex $u_{i,0}$ of $E_{i,0}$ and a vertex $u_{i,k}$ in $E_{i,k} \setminus E_{i,k-1}$ at every subsequent step. These are fixed label choices. In every convex realization their homogeneous representatives are independent: a supporting hyperplane for $E_{i,k-1}$ excludes the new vertex and contains all earlier chosen vertices. Thus the $R_0 \times (R_0 - 1)$ matrix T_i of these columns spans the homogeneous hyperplane of F_i .

Let ϕ_i be its cofactor functional, equivalently

$$\phi_i(w) = \det(T_i, w), \quad f_i(w) = \frac{\phi_i(w)}{\phi_i(q_i)}.$$

The denominator is nonzero in every realization because q_i is outside F_i . Since p_i, q_i lie on the same strict side of the supporting hyperplane, $f_i(p_i) > 0$ and $f_i(q_i) = 1$. The functionals f_i are independent: evaluation on q_j isolates the j -th coefficient of any linear relation among them.

Now change homogeneous representatives by the explicit positive rescalings

$$\hat{p}_i = \frac{p_i}{f_i(p_i)}, \quad \hat{q}_i = \frac{q_i}{f_i(q_i)} = q_i.$$

These are column rescalings, not ambient coordinate changes. They give

$$f_j(\hat{p}_i) = f_j(\hat{q}_i) = \delta_{ij}.$$

Moreover, the functional $\tau = \sum_i f_i$ is positive on every original vertex and therefore on the whole original polytope. All rescaled representatives lie in $\tau = 1$.

To see the Lawrence form, temporarily complete the independent f_i to dual coordinates. In these coordinates the rescaled matrix is $\begin{pmatrix} X & Y \\ I_N & I_N \end{pmatrix}$. The shear $(x, t) \mapsto (x - Yt, t)$ gives $\begin{pmatrix} X-Y & 0 \\ I_N & I_N \end{pmatrix}$. Full homogeneous rank implies $\text{rank}(X - Y) = r$, and part (1) shows that $X - Y$ realizes M . This establishes existence without yet specifying a rational choice of the temporary completion.

We can avoid that choice entirely. Set $\Delta_i = \hat{p}_i - \hat{q}_i$ and fix an ordered basis $J = (j_1, \dots, j_r)$ of M . The preceding Lawrence form shows that $\Delta_{j_1}, \dots, \Delta_{j_r}$ are independent in every realization. They span $\bigcap_i \ker f_i$, while the $\hat{q}_i$ are independent and complementary to that subspace. Hence the square matrix

$$H = (\Delta_{j_1}, \dots, \Delta_{j_r}, \hat{q}_1, \dots, \hat{q}_N)$$

is invertible everywhere. Apply the one ambient coordinate change H^{-1} to obtain

$$H^{-1}(\hat{p}_1, \dots, \hat{p}_N, \hat{q}_1, \dots, \hat{q}_N) = \begin{pmatrix} V' & 0 \\ I_N & I_N \end{pmatrix}, \quad V'_J = I_r.$$

The bottom coordinates of H^{-1} are exactly $f_1, \dots, f_N$, as is verified on every column of H . Thus its target affine chart has denominator τ , positive on the whole original polytope. The coordinate change is an allowed convex projectivity, and includes the effect of the temporary shear.

Every entry in this construction is rational over $\mathbb{Q}$. Its divisions use only $\phi_i(q_i)$, the positive numbers $f_i(p_i)$, and $\det H$, all nonzero at every real realization. Finally apply the fixed support-forest normalization to V' . Its row and positive column rescalings extend to an allowed block-diagonal convex projectivity as in part (3). We have obtained a global rational representative without any realization-dependent choices.

(3) *Equivalence classes.* Let a projectivity H take $\Lambda(V)$ to $\Lambda(W)$ with labels preserved. Since it preserves the facets F_i , its bottom-left block vanishes and its bottom-right block is diagonal. Since it preserves the labeled q_i , its top-right block vanishes as well. Thus, up to a common scalar,

$$H = \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_N), \quad w_i = Av_i/d_i.$$

The numbers d_i can be taken positive. A projectivity defined on a convex polytope has denominator of constant sign there, so its homogeneous vertex multipliers have one common sign, which we fix globally. Conversely any invertible A and positive diagonal D induce this labeled convex projective equivalence. The remaining freedom is exactly invertible row transformation and positive column scaling on V .

Statements (1)–(3) prove the bijection, and the rational reduction proves that it and its inverse have no real poles in the chosen global normalizations. $\square$

Proof of Theorem 1.2. Apply Proposition 5.1 to the loopless, coloopless oriented matroid supplied by Theorem 1.1. Its full Lawrence polytope has exactly the required projective realization space. The covectors determine its labeled face lattice, so the construction is effective. $\square$

For the construction with n_L line labels and K quadrilateral relations, substituting (4.2) into $d = r + N - 1$ gives dimension $d = n_L + 5K + 1$ and $2n_L + 8K$ vertices.

Projective equivalence is essential to this statement. Fixing an affine basis would leave parameters which the full projective quotient removes. Nor does the argument give a dimension bound: both the matroid rank and the number of Lawrence pairs grow with the input.

6. A CIRCLE AS A REALIZATION SPACE

We give one explicit circuit to illustrate the construction. The point is to produce the circle itself as the realization space, with no additional placement parameters. We do not attempt to minimize the rank or the number of elements.

For $x^2 + y^2 = 1$, set $u = x + 3$ and $v = y + 3$. The circle is polynomially isomorphic to the entire real locus of

$$(6.1) \quad u^2 + v^2 + 18 = 6(u + v) + 1.$$

Indeed, the equation is $(u - 3)^2 + (v - 3)^2 = 1$. Both u, v lie in $[2, 4]$, so we can use this simpler positive-coordinate encoding. Consider the eleven gates

$$\begin{array}{ll} \text{two} &= 1 + 1, & \text{three} &= \text{two} + 1, \\ \text{six} &= \text{three} + \text{three}, & \text{eighteen} &= \text{three} \cdot \text{six}, \\ a &= u \cdot u, & b &= v \cdot v, \\ c &= a + b, & L &= c + \text{eighteen}, \\ d &= u + v, & e &= \text{six} \cdot d, \\ R &= e + 1. \end{array}$$

The terminal equality $L = R$ is (6.1). Every circuit variable has value greater than 1. We order the clusters as follows: the initial cluster; the variable clusters for u, v and then the eleven gate outputs in the displayed

order; the eleven gate clusters in that same order; and the terminal equality cluster. In the displayed two-column array, the order proceeds by rows from left to right.

There are 13 variable clusters, seven addition clusters, four multiplication clusters, and 26 clusters altogether. The total number of local point occurrences is

$$3 + 13 \cdot 5 + (7 \cdot 5 - 3) + 4 \cdot 5 + 4 = 124.$$

The subtraction of 3 comes from the three additions with second input 1. Identifying 25 consecutive endpoints leaves 99 finite line labels, hence $n_L = 100$ including infinity. Removing the 26 local zero occurrences leaves 98 nonzero formal-symbol occurrences. Their distinct symbols are the unit, its negative, and the positive, negative and reciprocal symbols for every circuit variable: $2 + 3 \cdot 13 = 41$. Thus there are $98 - 41 = 57$ linking relations.

There are $1 + 2 \cdot 13 + 11 + 1 = 39$ local relations, and therefore $K = 39 + 57 = 96$. By (4.2) we obtain an oriented matroid of rank 98 on 484 elements. In particular,

$$\mathcal{R}_{\text{proj}}(M) \cong_{\mathbb{Q}} \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

Equalities such as $u = v$ at some circle points cause no extra point identifications: the two variable clusters have different local origins. This is why the same oriented matroid works on the entire circle.

For a concrete rational representative, take $(x, y) = (1, 0)$, so $u = 4, v = 3$, and evaluate the displayed circuit. The endpoint recurrence (2.1), the linking rule (2.2), and the coordinates (3.2) specify its full rational homogeneous matrix. The supplementary file `supplements/circle_construction.py`, distributed with the paper's source package, implements these formulas using exact rational arithmetic. It checks the relations, nondegeneracy conditions, endpoint identifications and all the construction counts at five specified rational circle points. Running it with `-export supplements` also writes `supplements/circle_lawrence_sparse.json`, the sparse rational homogeneous vertex matrix for $(x, y) = (1, 0)$. These finite checks make the representative reproducible; the full realization-space statement follows from Lemma 3.1 and Proposition 4.1. The full Lawrence polytope then has dimension 581 and 968 vertices, and its projective realization space is exactly the same circle. These are construction sizes, not minimal values.

7. THE RANK TRADEOFF

The construction removes two kinds of parameters in different ways. Endpoint gluing determines the translations of the arithmetic clusters by equations. Independent transverse directions allow one ambient projectivity to normalize the placements of all quadrilaterals. Neither operation replaces the input by a family with nontrivial fibers.

The price is visible in $\text{rank}(M) = K + 2$. Classical universality keeps the rank at three and permits stable equivalence. Here we obtain exact rational equivalence, but the rank cannot currently be bounded independently of the arithmetic circuit by this method. The growing rank is therefore the major tradeoff, rather than a secondary quantitative inconvenience.

Question 7.1. Does there exist an integer r_0 such that every nonempty basic semialgebraic set over $\mathbb{Q}$ defined by polynomial equations and strict inequalities is exactly rationally equivalent to the labeled projective realization space of an oriented matroid of rank at most r_0 ?

In particular, it is open whether one can take $r_0 = 3$. Merely choosing canonical placements in a classical planar construction does not answer the question: a chosen section of its auxiliary parameters need not be forced by the oriented matroid. A bounded-rank construction would have to eliminate those additional realizations by incidences, or use a different encoding whose projective quotient already has unique fibers.

The rational-coefficient hypothesis and the real-locus interpretation are also part of the result. We do not assert universality for arbitrary real coefficients, arbitrary Boolean combinations of inequalities, complex-scheme isomorphisms, or polytopes of fixed dimension. Within the stated class, however, the realization space itself is prescribed, rather than only its stable-equivalence class.

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